Dr. Abide Aşıkoğlu

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Delft University of Technology, Faculty of Civil Engineering & Geosciences, Department 3MD – Section Applied Mechanics, Building 23, Room 6.57, Stevinweg 1, 2628 CN Delft, The Netherlands.

EDUCATION

September 2018 – June 2024	**PhD,** University of Minho, ISISE, ARISE, Department of Civil Engineering, Guimarães, Portugal. **Title:** Performance-based design approach on masonry buildings by means of experimental and numerical pushover analysis. **Supervisors:** Prof. Dr. Paulo B. Lourenço, Assoc. Prof. Dr. Graça Vasconcelos.
September 2016 – July 2018	**MSc.,** Graduate School of Sciences, Department of Civil Engineering, Structural Mechanics, Anadolu University, Eskişehir, Türkiye. **Title:** Numerical Investigation on the Effectiveness of Seismic Retrofitting of a Historical Masonry Mosque. **Supervisor:** Prof. Dr. Özgür Avşar. High Honours, GPA: 3.71/4.00.
February – June 2017	**MSc. Visiting Student,** Structural Engineering, Middle East Technical University, Ankara, Türkiye.
September 2012 – July 2016	**BSc.,** Faculty of Engineering, Department of Civil Engineering, Anadolu University, Eskişehir, Türkiye. Honours, GPA: 3.10/4.00.
September 2014 – February 2015	**BSc.,** Faculty of Civil Engineering, Brno University of Technology, Brno, Czech Republic. Erasmus – European Union Student Exchange Program.

ACADEMIC POSITIONS

May 2024 – Present	**Postdoctoral Researcher,** Delft University of Technology, Faculty of Civil Engineering and Geosciences, Department of Materials, Mechanics, Management & Design (3MD), Section Applied Mechanics, Delft, The Netherlands. **Project:** Characterisation of Light Damage in Calcium-Silicate Block Masonry. **Role:** Development of semi-automated calibration and parametric analysis on calcium-silicate masonry walls to obtain fragility curves using machine learning methods. **Project Leaders:** Assist. Prof. Dr. Paul Korswagen, Prof. Dr. Jan Rots.

SCHOLARSHIPS AND AWARDS

November 2023 – February 2024
Individual PhD Scholarship, Exceptional Scholarship due to COVID-19, Fundação para a Ciência e a Tecnologia (FCT), Portugal.
Grant NO: COVID/BD/153438/2023.
DOI: https://doi.org/10.54499/COVID/BD/153438/2023

September 2019 – October 2023
Individual PhD Scholarship, Fundação para a Ciência e a Tecnologia (FCT), Portugal.
Grant NO: SFRH/BD/143949/2019.

June 2023
Inaugural Women in Masonry Scholarship (The Masonry Society), sponsored by Ece Erdoğmuş and Brian Skourup, to attend the 14th NAMC between June 11thand 14th, 2023.

March 2022
Editor's Choice Article, MDPI Buildings.
Title: Overview on the Nonlinear Static Procedures and Performance-Based Approach on Modern Unreinforced Masonry Buildings with Structural Irregularity, DOI: https://doi.org/10.3390/buildings11040147

October 2020
MSc Thesis Award, Eskişehir Technical University, Eskişehir, Türkiye.

July 2016 – July 2018
Individual MSc Scholarship, Turkish Republic of Northern Cyprus, Ministry of National Education and Culture.

May 2016
2nd Prize, Anadolu University Faculty of Engineering National Project Fair, Eskişehir, Türkiye.
Poster Presentation.

FUNDING

February – June 2016
Industry-oriented undergraduate thesis program, 2241A
Project Leader, "Düşey Yönde Süreksiz Olan Kolonun Kat Kesme Oranının Bina Sismik Performansına Etkisi" (Effect of Story Shear Force Ratio of the Discontinuous Column on the Seismic Performance of RC Building).
Supervisor: Prof. Dr. Özgür Avşar
Funding No: 1139B411503049
Funding Entity: The Scientific and Technological Research Council of Türkiye (TÜBİTAK).

RESEARCH

International Collaboration
Multihazard Assessment of the Adıyaman Grand Mosque
Collaboration Team: Abide Aşıkoğlu, Özgür Avşar, Fikret Kuran, Paul Korswagen, Alessandro De Iasio, Giovanni Fabbrocino, Bahman Ghiassi.
November 2025 – Present.

Industry-oriented Research	**Duurzaam Herstel (DZH)** **Client:** Instituut Mijnbouwschade Groningen (IMG) **Principal Investigator:** Paul Korswagen, Jan Rots. **Collaboration Team:** Paul Korswagen, Michele Longo, **Abide Aşıkoğlu,** Alfonso Prosperi. **Duration:** June 2025 – Present.
Blind Prediction Competition Participant	**ERIES Project: Response Evaluation of masonry VAULTs under pseudo-dynamic loadings (REVAULTs)** **Collaboration Team: Abide Aşıkoğlu,** Amir Ghezelbash, Eduarda Vila- Chã, Elesban Nochebuena Mora. 2025
Blind Prediction Competition Participant	**Shaking Table Tests on 29 Masonry Model Structures** **Collaboration Team:** Amir Ghezelbash, **Abide Aşıkoğlu,** Antonio Maria D'Altri, Satyadhrik Sharma, Francesco Messali **Winner** 2025
PhD Topic	**PUMA – Experimental and numerical pushover analysis of masonry buildings** **ISISE Principal Investigators:** Paulo Lourenço, Graça Vasconcelos. **ID:** POCI-01-0145-FEDER-029010 **Budget:** Global: 239670,62€ / ISISE-UM: 204558,12€ **Funding Entity:** COMPETE2020/FCT **Duration:** 2018 – 2022.
Blind Prediction Competition Participant	**The Adjacent Interacting Masonry Structures (AIMS)** subproject from the Seismology and Earthquake Engineering Research Infrastructure Alliance for Europe (SERA). **Collaboration Team: Abide Aşıkoğlu,** Antonio Romanazzi, Fabio Solarino, Jennifer D'Anna, Maria Pia Ciocci, Nicoletta Bianchini, Rafael Ramirez Alvarez de Lara. 2019 – 2022.
Erasmus+ Traineeship Program Structural Engineering	**University of Minho,** Guimarães, Portugal. Development of a finite element analysis of a historical masonry mosque within the scope of the MSc Thesis. **Supervisor:** Paulo B. Lourenço, **Mentor:** Luis C. Silva. September 2017 – January 2018.
Expertise and Interests	Laboratory Testing, Experimental Analysis, Structural and Earthquake Engineering, Augmented Intelligence, Data Mining, Structural Mechanics, Masonry Structures, Structural Dynamics, Finite Element Analysis, Discrete Macro-Element Modelling, Equivalent Frame Modelling, Earthquake-Based Design, and Reinforced Concrete.

JOURNAL PUBLICATIONS

Ghezelbash A., **Aşıkoğlu A.**, D'Altri A.M., Sharma S., Messali F., (2026), Blind prediction of the dynamic behavior of 3D-printed masonry-like structures under shake-table loading via a high-fidelity numerical modeling approach Bull Earthquake Eng, https://doi.org/10.1007/s10518-026-02565-y

Aşıkoğlu A., Vasconcelos G., Barontini A., Lourenço P., (2025), Quasi-Static Test on Half-Scale Modern Unreinforced Masonry Building with Plan Irregularity. Bull Earthquake Eng, https://doi.org/10.1007/s10518-025-02275-x

Bianchini, N., Ciocci, M.P., Solarino, F., et al., **Aşıkoğlu A.,** (2024), Influence of wall-to-floor connections and pounding on pre- and post-diction simulations of a masonry building aggregate tested on a shaking table. Bull Earthquake Eng, https://doi.org/10.1007/s10518-023-01641-x

Aşıkoğlu A., D'Anna J., Ramirez R., et al., (2024), Simulation of blind pre-diction and post-diction shaking table tests on a masonry aggregate using a continuum modelling approach, Bull Earthquake Eng, https://doi.org/10.1007/s10518-023-01666-2

Tomić, I., A. Penna, M. DeJong, et al., **Aşıkoğlu A.**, (2024), Shake-table testing of a stone masonry building aggregate: Overview of blind prediction study. Bull Earthquake Eng, https://doi.org/10.1007/s10518-022-01582- x

Aşıkoğlu A., Avşar Ö., (2022), Investigation of Drift-based Damage Limit States for Historical Masonry Structures, International Journal of Architectural Heritage, 17:9, 1571-1589, https://doi.org/10.1080/15583058.2022.2053612

Aşıkoğlu A, Vasconcelos G, Lourenço P.B., (2021), Overview on the Nonlinear Static Procedures and Performance-Based Approach on Modern Unreinforced Masonry Buildings with Structural Irregularity. Buildings. 11(4):147. https://doi.org/10.3390/buildings11040147

Aşıkoğlu A., Vasconcelos G., Lourenço P.B., Pantò B, (2020), Pushover analysis of unreinforced irregular masonry buildings: Lessons from different modeling approaches, Engineering Structures, Volume 218, 110830, ISSN 0141-0296, https://doi.org/10.1016/j.engstruct.2020.110830

Aşıkoğlu, A., Avşar, Ö., Lourenço, P.B., Silva L.C. (2019) Effectiveness of seismic retrofitting of a historical masonry structure: Kütahya Kurşunlu Mosque, Turkey. Bull Earthquake Eng 17, 3365–3395. https://doi.org/10.1007/s10518-019-00603-6

CONFERENCE PROCEEDINGS

Aşıkoğlu A., Longo M., Korswagen P., (2026). "Lessons learned from damage mechanism verification beyond force, displacement and stiffness matching in numerical modelling of masonry structures", 11th International Masonry Conference, July 12 – 15, Lübeck, Germany. (Oral Presentation)

Ghezelbash A., **Aşıkoğlu A.**, D'Altri A.M., Sharma S., Messali F., (2026) "High-fidelity numerical modeling of the dynamic behavior of 3D-printed masonry-like sub-structures under shake-table loading", 10th European Congress on Computational Methods in Applied Sciences and Engineering, 19th – 24th July 2026, Munich, Germany. (Oral Presentation)

Aşıkoğlu A., Dhiandra Riza A., Korswagen P., Kuran F., Avşar Ö., (2025). "Domino Collapse in Urban Settings: An 19th Century Mosque Caused to Collapse During the 2023 Kahramanmaraş Earthquake Sequence", 14th International Conference on Structural Analysis of Historical Constructions (SAHC 2025), September 15th – 17th, 2025, Lausanne, Switzerland. (Oral Presentation)

Aşıkoğlu A., Korswagen P., Rots J., (2025). "Optimizing Numerical Modelling for In-Plane Response of Clay and Calcium Silicate Masonry through Data-Informed Semi-Automation", 15th Canadian Masonry Symposium, June 2nd – 5th, 2025, Ottawa, Canada. (Oral Presentation).

Aşıkoğlu A., Vasconcelos G., Lourenço P.B., (2024). "Decoding Seismic Response in Modern Unreinforced Masonry: Experimental Insights on Plan Irregularity", 18th World Conference on Earthquake Engineering (WCEE 2024), June 30th – July 5th, 2023, Milan, Italy. (Oral Presentation)

Aşıkoğlu A., Avşar Ö., (2024). "Drift-based Damage Limits in Seismic Performance Assessment of Historical Masonry Structures", 18th World Conference on Earthquake Engineering (WCEE 2024), June 30th – July 5th, 2023, Milan, Italy. (Poster Presentation)

Aşıkoğlu A., Vasconcelos G., Lourenço P.B., (2004). "A critical overview of challenges in quasi-static testing of a two-story unreinforced masonry building", 13º Congresso Nacional de Sismologia e Engenharia Sísmica (Sísmica 2024), April 17th – 19th 2024, Guimarães, Portugal. (Oral Presentation)

Aşıkoğlu A., Barontini A., Savalle N., Vasconcelos, G., & Lourenço & P. B., (2023). "Experimental analysis of unreinforced masonry buildings through quasi-static test: A half-scale two-story modern masonry building", 14th North American Masonry Conference (14th NAMC), June 11th – 14th 2023, Omaha, Nebraska, the USA. (Oral Presentation)

Aşıkoğlu A., Vasconcelos, G., & Lourenço P. B., (2022). "Cyclic Quasi-Static Testing of a Half-Scale, Two-Story Unreinforced Masonry Building with Structural Irregularity", 12th National Congress on Seismology and Earthquake Engineering (JPEE 2022), November 9th – 11th 2022, Lisbon, Portugal. (Oral Presentation)

Tomić, I., A. Penna, M. DeJong, et al., (2022), "Seismic testing of adjacent interacting masonry structures-shake table test and blind prediction competition", 3rd European Conference on Earthquake Engineering and Seismology, 4-9 September 2022, Bucharest, Romania.

Aşıkoğlu A., Del Re, A., Vasconcelos, G., Lourenço & P. B., (2021). "Material characterization and quasi-static test on a half-scale two-story URM building", 12th National Congress on Experimental Mechanics, September 29th – October 1st, 2021, Monte Real, Leiria. (Oral Presentation)

Aşıkoğlu A., Silva L. C., Avşar Ö., Lourenço P.B., (2020). "Assessment of the Seismic Retrofitting of a Historical Masonry Mosque by Means of Nonlinear Dynamic Analysis", 12th International Conference on Structural Analysis of Historical Constructions (SAHC 2020), 16-18 September 2020, Barcelona, Spain. (Pre-recorded Presentation)

Aşıkoğlu A., Vasconcelos G., Lourenço P.B., "Pushover Analysis of an Asymmetric Unreinforced Masonry Building; Experimental and Numerical Results", 17th International Brick and Block Masonry Conference: From Historical to Sustainable Masonry, 5-8 July 2020, Krakow, Poland. (Online Presentation)

Aşıkoğlu A., Avşar Ö., Lourenço P.B., Silva L.C., "Seismic Assessment of Historical Kütahya Kursunlu Mosque", 4th International Conference on Preservation, Maintenance and Rehabilitation of Historic Buildings and Structures (REHAB 2019), 17-19 July 2019, Guimarães, Portugal. (Oral Presentation)

Aşıkoğlu A., Vasconcelos G., Lourenço P.B., Panto B., Marques R., "Nonlinear Static Analysis of an Asymmetric Unreinforced Masonry Building Using Macro-Element Modelling Approach", Congress on Numerical Methods in Engineering (CMN 2019), 1-3 July 2019, Guimarães, Portugal. (Poster and Oral Presentation)

Aşıkoğlu A., Avşar Ö., Lourenço P.B., Silva L.C., Kaplan O., Karanikoloudis G., "Finite Element Modeling and Operational Modal Analysis of a Historical Masonry Mosque", 7th International Conference on Computational Methods in Structural Dynamics and Earthquake Engineering (COMPDYN 2019), 24-26 June 2019, Crete, Greece. (Oral Presentation)

Aşıkoğlu A., Avşar Ö., "A Case Study on the Adequacy of TEC 2007 Requirements for the Column Discontinuity", 4th International Conference on Earthquake Engineering and Seismology (4ICEES), 11 - 13 October 2017, Eskişehir, Türkiye. (In Turkish) (Oral Presentation)

Aşıkoğlu A., Avşar Ö., Kuran F., Bayülke N., "Dynamic response of retrofitted Kütahya Merkez Kurşunlu Mosque", 3rd International Conference on Protection of Historical Constructions (PROHITECH'17), Paper No. 82, 12 - 15 July 2017, Lisbon, Portugal. (Oral Presentation)

DATASETS

Aşıkoğlu, A., Barontini, A., Vasconcelos, G., & Lourenço, P. (2025). Experimental Dataset on Cyclic Quasi-Static Testing of a Half-Scale Modern Unreinforced Masonry Building with Plan Irregularity [Data set]. Zenodo. https://doi.org/10.5281/zenodo.15096909

SUPERVISION & MENTORING

MSc. Thesis – Co-supervisor
Aldy Riza Dhiandra, (2025). "Heritage Amidst Modern Failures; Assessing the Impact of Nearby Building Collapse on Historical Masonry Mosque in an Urban Setting: Case Study from the 2023 Kahramanmaraş Earthquake Sequence", Delft University of Technology.

MSc. Thesis – Mentor
Antoniya Ivanova Ivanova (2022). "Experimental Behaviour of a Structural Masonry Building", University of Minho, (Completed)

MSc. Thesis – Mentor
Alessandro Del Re (2020). "On the Seismic Behaviour of Unreinforced Masonry Buildings with Complex Geometry: Numerical Modelling and Experimental Characterisation", Alma Mater Studiorum Universitá di Bologna & University of Minho. (Completed)

EDUCATION

CIEM5210 Applied Mechanics of Structures, Computational Modelling of Structures Unit, Teaching Assistant, TU Delft, 2025, 2026.

CT3280-19 Bend and Break – Masonry Workshop. Teaching Assistant, TU Delft, October-December 2025. (Bachelor Course)

CIEM 5307 Forensic Structural Engineering. Invited Lecture, "Numerical Modelling for Forensic Structural Engineering: Masonry Structures", TU Delft, October 2025. (Master Course)

LABORATORY EXPERIENCE

- Determination of the density of the clay brick unit (EN 1015-10, 1990),
- Compression test on masonry units (EN 771-1, 2000),
- Flow table test on fresh mortar (EN 1015-3, 1999),
- Flexural test on mortar (EN 1015-11, 2007),
- Compression test on mortar (EN 1015-11, 2007),
- Compression test on concrete (EN 12390-3, 2019).
- Uniaxial compression test (EN 1052-1, 1999),
- Flexural strength test (EN 1052-2, 1999),
- Initial shear test (EN 1052-3, 2002),
- Diagonal compression test (ASTM E 519 - 02, 2002),
- Ambient Vibration Measurements on a half-scale building,
- Digital Image Correlation application on a façade of a half-scale building,
- Cyclic quasi-static test on a half-scale two-story URM building.

EDITORIAL & REVIEWING ACTIVITIES

Guest Editor

Aşıkoğlu, A., Parisse, F., & Bianchini, N. (Guest Editors). *Symmetry (MDPI)*, Special Issue: "Structural Mechanics of Regularity and Irregularity in Masonry Constructions," 2026, ISSN 2073-8994; Section: Engineering and Materials. https://www.mdpi.com/journal/symmetry/special_issues/MQJVE978UI

Book Reviews

Yurdakul, Ö. Matrix structural analysis by examples: Solutions for advanced structural mechanics. University of Pardubice, Pardubice, Czech Republic, 2022, ISBN: 978-80-7560-426-2.

Peer Reviews

Journal of Nondestructive Evaluation, *ISSN: 0195-9298*. January 2026.
Infrastructures, *ISSN: 2412-3811*. September 2025.
Bulletin of Earthquake Engineering, *ISSN: 1573-1456*. August 2025; May 2026.
International Journal of Architectural Heritage, *ISSN: 1558-3066*. October 2024.
Structures, *ISSN: 2352-0124*. May 2025. March 2023.
Arabian Journal of Science and Engineering, *ISSN: 1319-8025*. February 2023.
Scientia Iranica, *ISSN: 2345-3605*. July 2020.

SEMINARS, WEBINARS, AND INVITED LECTURES

Seminars / Webinars

Presenter, "Exploring Nonlinear Analysis of Unreinforced Masonry Buildings: Insights from Various Modelling Approaches", Computational Modelling Workshop on Structural Engineering, Dokuz Eylül University, 16th March 2024. (Online)

Presenter and organiser, "Experimental Pushover Analysis of a URM Building with Plan Irregularity", Recent advances on structural analysis of masonry, University of Minho, 14th of July 2022. (Webinar)
https://www.youtube.com/watch?v=7EtgYrwgMZI

Presenter and organiser, "Basic application of Structural Engineering in DIANA: Masonry Buildings", PDEC Software Series, University of Minho, 12th of April 2022. (Webinar)

Presenter, "Numerical simulation of quasi-static test on a half-scale URM building", 5th PDEC Workshop, University of Minho, 3rd of December 2021. (Seminar)

Presenter and organiser, "Introduction to 3DMacro & a case study", PDEC Software Series, University of Minho, 30th of November 2021. (Webinar)

Invited Lectures

Speaker, "Experimental mini-setup for facades subjected to ground deformations" – Invited lunch lecture at GEMMA project meeting, 16th of December 2025. (In-person)

Speaker, "An Investigation on the Effectiveness of Seismic Retrofitting of a Historical Masonry Mosque", Deprem Güçlendirme Derneği (DEGÜDER) – The Earthquake Strengthening Association, 14th of December 2022. (Webinar) (In Turkish)
https://www.youtube.com/watch?v=xmQwLovo5s4&t=754s

Speaker, Women in Civil Engineering (CE), ESTÜYAP, Eskişehir Technical University, 8th of March 2021. A webinar to commemorate International Women's Day. (In Turkish)
https://www.youtube.com/watch?v=fcYZBeJSL0U&t=1940s

SKILLS

Language

Turkish (native proficiency), English (professional working proficiency), French (elementary proficiency), Portuguese (elementary proficiency).

Software

3DMacro, 3Muri, AutoCAD 2D and 3D Drawing, DIANA FEA, ARTeMIS Modal, MathCAD, MATLAB, XTRACT, STA4CAD, SAP2000, Microsoft Office.

Transferable

Communication – Meetings, dissemination, oral presentations.

Teamwork – Had the opportunity to work with various researchers with different backgrounds.

Organisation – Community involvement.

Delft University of Technology: Member of the university-wide Postdoc Network Committee, coordinating communications and outreach across the postdoctoral community. Contributed to disseminating information on researcher rights and institutional policies through the network's communication channels, supporting the representation and well-being of postdoctoral researchers across all faculties.

University of Minho: Member of the PhD student committee in Civil Engineering, contributing to the organisation of initiatives aimed at supporting the doctoral community, including "Mind the PhD" to raise awareness of mental health challenges during doctoral training, "Women in PDEC" and "International Women's Day" to promote gender equality and inclusivity in research, and "PubDEC" to foster informal dissemination of research among peers.

INTERESTS

Community

TU Delft Postdoc Network Committee Member, Delft University of Technology, September 2025 – Present.

Representative in Portugal, European Union Turkish Cypriot Student Federation (NGO), September 2021 – March 2024.

PhD Student Committee Member, Department of Civil Engineering, University of Minho, June 2020 – July 2022.

Heritage Youth Ambassador, Technical Committee on Cultural Heritage in Cyprus, UNDP Cyprus, 2019 – 2021.

Personal

Reading, philosophy, psychology, volleyball, Pilates, travelling, painting, handcrafting, baking.

SCIENTIFIC SOCIETIES

September 2018 – June 2024	**PhD. Candidate/Researcher,** Institute for Sustainability and Innovation in Structural Engineering (ISISE)
March 2023 – March 2024	**Student Membership,** The Masonry Society (TMS)
November 2020 – November 2021	**Student Membership,** Earthquake Engineering Research Institute.

PERSONAL INFORMATION

Name and surname	Abide Aşıkoğlu
Date of birth and place	21/08/1994, Famagusta, North Cyprus
Webpages	Abide Aşıkoğlu - ResearchGate Abide Aşıkoğlu - Google Scholar Abide Aşıkoğlu | LinkedIn TU Delft Staff - A. (Abide) Ashikoglu Aşıkoğlu, Abide - Scopus